Yuan Zhang

Email:31415926harper@gmail.com
My Academic Page:https://harper-zh.github.io/
Phone: +86 15713200193

EDUCATION

Southeast University Sept. 2023 - Jun. 2026 (expected)
M.Arch, Lab of Architectural Algorithm and Application Score: 87.23/100

Southeast University Sept. 2018 - Jun. 2023
B.Arch GPA: 3.77/4.0

SKILLS

- **Computational Design:** Rhino, Grasshopper, Geometry Algorithms
- **Robotics & Fabrication:** Path Planning, Simulation, 3D printing, Timber Assembly
- **Programming:** Python, JavaScript, TypeScript, Vue.js, HTML, CSS
- **AI & Data Processing:** Machine Learning (Scikit-learn, PyTorch), Deep Learning (VAE, CNN Basic), Data Processing (NumPy, Pandas), Data Visualization (Matplotlib, Seaborn)
- **Language:** English (IELTS 7.5), German (Preliminary)

PUBLICATIONS

- [1] Yuan Zhang, Hao Hua. Light Redistribution in Variable 3D-Printed Thermoplastic Geometries: Automated optics experiments and machine learning. Presented at the eCAADe, Ankara, Turkey, Sept. 2025. Accepted for publication in conference proceedings.

RESEARCH EXPERIENCE

Parametric Design and Robotic Construction of Timber Floor Systems Apr. 2025 - Sept. 2025

Research Intern | Supervisor: Prof. Kathrin Dörfler, TU Munich

- Contributed to the parametric design of timber floor frameworks and the development of a structural integration tool, enhancing structural integration and construction sequence visualization through Python scripts.
- Simulated robotic construction processes by modeling the building environment and visualizing robotic motion and assembly process in Rhino, based on the COMPAS framework and ROS.
- Participated in on-site construction, completing robotic fabrication and validation of two timber floor slabs (6m × 2m).

Light Redistribution in Variable 3D-Printed Thermoplastic Facade Dec. 2024 - Present

Master Thesis | Supervisor: Prof. Hao Hua | Presented at eCAADe 2025 (see Publication [1])

- Researched spatial light redistribution of daylight passing through 3D-printed thermoplastic façades (3DPTF) and its relationship with geometric parameters. Based on this, proposed a systematic approach combining automated robotic experiments and machine learning to support indoor lighting improvement and façade design optimization.
- Conducted physical experiments to collect daylight distribution data under varying geometric parameters, establishing a dataset of optical behaviour for 3DPTF specimens.
- Proposed a classification and description method for optical behaviour features and and used machine learning to establish predictive relationships between geometric parameters and lighting performance.
- Evaluated the applicability of optical behaviour features in architectural design.

Design and Digital Fabrication of Baimaofang: An Interactive Sculpture Feb. 2023 - May 2023

Bachelor Thesis | Supervisor: Prof. Li Li

- Researched interaction methods and digital fabrication techniques for a freeform interactive concrete sculpture located by the river.

- Optimized freeform surfaces into ruled surfaces for efficient fabrication, designed concrete-to-metal connections, customized robotic hot wire cutting for formwork, and conducted concrete casting test with foam molds.

PROJECTS

3D-Printed Urban Furnitures for Dingshu Town Aug. 2024 - Oct. 2024

- Responsible for developing path planning program and robotic motion simulation, generating 45 programms sucessfully.
- Fine-tuned printing parameters and oversaw on-site printing operations, controlling color and final product quality. Successfully printed 15 large-scale urban furniture, each 3.6 meters in length.

EXTRACURRICULAR EXPERIENCES

VITAMIN – Female-Centric AI Fortune Telling Application Dec. 2025

AI Product Engineer | Website Link: https://vita-me.vercel.app/

- Product Definition: Addressed the lack of emotional resonance and modern feminism perspectives in traditional fortune-telling tools, responsible for the entire process from user insights and feature planning to front-end interaction and deployment.
- Interaction Strategy: Implemented a Perceptual Latency Elimination strategy using layered asynchronous rendering and typewriter effects, reducing user-perceived response time to 2s.
- Impact: Delivered a high-fidelity MVP, achieving over 90% user satisfaction in closed beta testing, specifically praised for interactive details and niche market positioning.

TEACHING EXPERIENCES

Teaching Assistant, Computational Design and Digital Fabrication Sept. 2025 - Dec. 2025

- Supported course instruction, supervised fabrication sessions, assisted students with project proposal and Python teaching

AWARDS

- Southeast University Graduate Second-Class Academic Scholarship 2023
- 2nd Prize in the Zijin Award Design Competition 2022
- Jiangsu Province Construction Scholarship 2021
- 2nd Prize in Undergraduate Thesis on History and Theory from School of Architecture SEU 2021
- 1st Prize in the Southeast University Innovation Competition 2019

LEADERSHIP EXPERIENCE

- President of the Student Union Jun. 2021 - Jun. 2022
- Head of Student Services Center Jun. 2020 - Jun. 2021